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SDC Hackathon

Technical Tools Declaration

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Declare every API, cloud service, AI tool, dataset, model, and sponsor resource your team used to build the project. Complete every section; where a category does not apply, tick "No" or write "None" – do not leave it blank. Submit as a single PDF document named "SDC2026_TeamName_TechDeclaration.pdf" together with your project submission. Undeclared third-party resources discovered at judging may affect your score or eligibility.

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1. Team & Project Overview

Team Name	CodeBlack
Project / Solution Name	Mustaqbal AI
Name (Leader)	Rehan Anjum
Email & Phone (Leader)	rehanjum9@gmail.com, +60 17-623 3902
University Affiliation(s)	UCSI University, Malaysia
Live Prototype URL	https://mustaqbal-ai.replit.app
Code Repository URL (public / access granted to judges)	https://github.com/xlBABA/CodeBlack_SDC

2. Architecture & Technology Stack

2.1 System Architecture Overview

Describe how the components declared in this document connect end-to-end (e.g., user interface → backend → AI model/API → database → cloud hosting). Attach a system block diagram as an appendix if available.

[Maximum 200 words]

Mustaqbal AI is one web application hosted on Replit. A React 19 and TypeScript single-page frontend (built with Vite) is served by a FastAPI backend. The backend keeps student profiles in SQLite and reads cached, versioned data files: 30 roles derived from O*NET, 20 skills, learning resources, project ideas and GitHub trend snapshots. A student types skills or uploads a CV PDF. The PDF text is extracted locally and sent to Z.ai GLM-5.3 on a RunPod Public Endpoint, which drafts an editable profile; the student must review and confirm it before matching, and a timeout falls back to manual entry. Matching is a deterministic weighted scoring engine with written rules, so every score has an explanation. When RunPod credentials are set, skill names are embedded by all-MiniLM-L6-v2 on a RunPod Serverless GPU endpoint, so related skills earn partial credit and free-text skills can be interpreted. If that endpoint fails, matching falls back to exact skill matching and every result is labelled degraded. Scenario simulation adjusts skill weights using GitHub Search trend snapshots that were collected offline and cached; the app does not query GitHub live. Roadmaps, project recommendations and SDG links come from the same cached data.

2.2 Languages, Frameworks & Libraries

List the main programming languages, frameworks, and third-party libraries that your team did not create. Open-source use is permitted provided it is declared and properly licensed/attributed.

Name & Version	Purpose in Project	License
Python 3.11 (RunPod worker), 3.12 (backend target)	Backend and AI worker language	Python Software Foundation License (language; not read from package metadata)
TypeScript 6.0.3	Frontend language	Apache-2.0
FastAPI 0.156.0	Backend web API	MIT
uvicorn[standard] 0.54.0	ASGI server	BSD-3-Clause (extras not checked)
Pydantic 2.14.1, pydantic-settings 2.16.1	Validation, schemas, typed configuration	MIT
SQLAlchemy 2.1.5	SQLite persistence of profiles	MIT

Name & Version	Purpose in Project	License
httpx 0.28.1, python-dotenv 1.2.4	GLM-5.3 client; environment files	BSD-3-Clause
pypdf 6.21.2	Local CV PDF text extraction	BSD-3-Clause
python-multipart 0.0.32	CV file upload	Apache-2.0
runpod SDK 1.12.0	Serverless handler in the RunPod worker	MIT
sentence-transformers 6.2.0	Runs all-MiniLM-L6-v2 in the worker	Apache-2.0
torch 2.14.1	Tensor runtime in the worker	Apache-2.0 AND Apache-2.0 WITH LLVM-exception AND BSD-2-Clause AND BSD-3-Clause AND BSL-1.0 AND MIT
transformers 5.20.2	Model loading in the worker	Apache 2.0 License (as written in metadata)
React and react-dom 19.3.1	Frontend UI	MIT
react-router 8.5.0	Frontend routing	MIT
Vite 8.3.5, @vitejs/plugin-react 6.1.2	Frontend build tooling	MIT
Tailwind CSS and @tailwindcss/vite 4.3.3	Styling	MIT
radix-ui 1.8.1	Accessible UI primitives	MIT
lucide-react 1.61.0	Icons	ISC
class-variance-authority 0.7.1	Component variants	Apache-2.0
clsx 2.1.1, tailwind-merge 3.7.0, lenis 1.3.27	Class names, class merging, smooth scrolling	MIT
GSAP 3.15.0, @gsap/react 2.1.2	Frontend animation	GSAP Standard No-Charge License (free, not OSI open source)
pytest 9.1.1, oxlint 1.89.0, @types/react, @types/react-dom, @types/node	Testing, linting, type definitions (development only)	MIT

3. APIs & Cloud Services

3.1 External APIs

List every external API your solution calls (GenAI, maps, payments, messaging, data feeds, etc.). Do not include API keys or secrets in this document.

API & Provider	Purpose in Project	Access Tier (free / paid / sponsor-provided)
RunPod Serverless endpoint f0phohhegtydji (RunPod)	Embeddings from all-MiniLM-L6-v2 for skill similarity and free-text skill interpretation	Sponsor-provided (RunPod credits)
Z.ai GLM-5.3 via RunPod Public Endpoint b1dd8hy2hqo1nb (RunPod)	CV PDF text to an editable draft profile (education, skills, projects, experience, certifications)	Sponsor-provided (RunPod credits)
GitHub Search API (GitHub)	Repository counts for trend snapshots; collected offline and cached, not called live by the app	Free

3.2 Cloud Platforms & Deployment

List every cloud service used for hosting, storage, databases, authentication, or compute, and indicate how the prototype is deployed.

Service & Provider	Purpose in Project	Plan / Region
Replit (Replit Inc.)	Hosts the live app	Replit Core, Replit Deployments, default region
RunPod Serverless (RunPod)	GPU endpoint for embeddings; GLM-5.3 is reached through RunPod's Public Endpoint	Serverless, 24 GB GPU, any available region (auto-assigned)
SQLite file on the Replit deployment	Stores student profiles	Part of the Replit deployment; persistence across restarts not verified

4. AI Tools, Models & Datasets

4.1 AI Models Used in the Solution

List every AI model embedded in or called by your application, including pre-trained, fine-tuned, and self-trained models.

Model & Version	Source / Provider	Usage (pre-trained / fine-tuned / self-trained)	License / Terms
sentence-transformers/all-MiniLM-L6-v2 (384 dimensions; snapshot 1110a243fdf4706b3f48f1d95db1a4f5529b4d41)	Hugging Face (sentence-transformers); served on the team's RunPod Serverless endpoint	Pre-trained, used as is, not fine-tuned	Apache-2.0 (model card)
GLM-5.3	Z.ai, via RunPod sponsor Public Endpoint (b1dd8hy2hqo1nb)	Pre-trained, used as is; only for CV PDF text to an editable profile draft, never for scoring	Accessed via RunPod Public Endpoint under RunPod terms; model license per Z.ai's model card

4.2 Datasets

List all datasets used for training, fine-tuning, evaluation, or in the running application. State clearly if a dataset contains personal or sensitive data.

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
O*NET occupation data (30 occupations; importance scaled 1 to 5 by the team)	https://www.onetonline.org (U.S. Department of Labor, Employment and Training Administration)	Public	CC BY 4.0	N
GitHub repository-count snapshots (12 skills, retrieved 2026-10-21)	https://docs.github.com/en/rest/search/search#search-repositories	Public, collected by the team through the API	GitHub Terms of Service	N
Learning resources (16 links)	Third-party free courses and documentation. We only link to them and do not copy or redistribute them	Public (linked)	Not redistributed; each provider's own terms	N

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
Project dataset links (17 project ideas): 8 UCI ML Repository datasets and World Bank World Development Indicators; tutorials from Onshape, GnuCash, QGIS	https://archive.ics.uci.edu and the World Bank Data Catalog. Linked, never copied into the app	Public (linked)	CC BY 4.0	N
Synthetic student profiles and 52 fictional skill-name pairs for threshold calibration	Written by the team for tests and demos	Synthetic	Team-created	N

4.3 AI-Assisted Development Tools

Tools used to help write code, design, content, or documentation (e.g., chat assistants, code completion tools, image generators). Declaring these tools does not reduce your score; undeclared use may.

Did your team use any generative-AI tool to help create code, design, text, or other project material?

☑ Yes ☐ No

If yes, list them below:

Tool & Version	Used For	Extent of Use (minor / moderate / major)
Claude Code, Claude Sonnet 5.5 (CLI 2.1.306)	Hussain: matcher embeddings, calibration, RunPod worker and client, reviews, docs	Major
Claude chat (claude.ai), Claude Opus 5.5	Hussain: original matcher core and backend adapter	Major
ChatGPT (version not stated)	Rehan: planning, tech stack, code review, troubleshooting, deployment guidance	Moderate
Replit Agent (version not stated)	Rehan: CV extraction timeout investigation, FastAPI improvements, diagnostics, implementation, automated testing	Major
Claude Code, Claude Sonnet 5 then Opus 5.5	Jawad: source register, role and skill data, SDG mapping, trend feed, roadmaps, projects, tests, link checks	Major
Claude chat (claude.ai)	Jawad: planning, reviewing pushed code, web searches for course leads, drafting prompts; wrote no repo files	Minor
Claude Code (version not stated)	Mohammad: frontend development	Major
OpenAI Codex (version not stated)	Mohammad: frontend development	Not stated by member
Replit Agent (version not stated)	Mohammad: frontend development	Not stated by member

Briefly describe how your team reviewed, tested, and took responsibility for AI-generated output:

[Maximum 100 words]

AI output was reviewed through tests, diffs, live checks and cross-review, not by reading every line. Tests: 178 matcher and worker tests (HTTP mocked) and the backend regression suite. Diffs were read before commits, and members reviewed each other's changes at integration. The matcher was run against the live RunPod endpoint with 10 fictional students over 30 roles, and 300 explanations were checked automatically; wrong mappings were fixed or reported. AI-assisted work is logged in docs/AI_USAGE_LOG.md. A full line-by-line review was not done for all generated code.

5. Sponsor-Provided Resources

Declare any official SDC sponsor-provided GenAI API, cloud credits, tools, mentoring, or data that your team used.

Did your project use any official SDC sponsor-provided resource?

☑️ Yes ☐ No

If yes, list them below:

Sponsor	Resource (API / credits / platform / data)	How It Was Used in the Project
RunPod	$25 credits (balance $24.67 of $25 on 25 Oct 2026, 01:37); Serverless GPU endpoint; GLM-5.3 Public Endpoint	Credits fund both the embedding endpoint (skill similarity) and GLM-5.3 (CV extraction)
Replit	Replit Core plan via SDC sponsor voucher (one per participant)	Hosting and deployment of the live app

Was the sponsor resource essential to the core functionality of the prototype, or could it be replaced? Explain briefly:

[Maximum 100 words]

RunPod is replaceable for the core scoring but essential for the AI features as built. Matching, ranking and explanations are deterministic and run without it; if the embedding endpoint is down, results fall back to exact skill matching and are labelled degraded. RunPod adds semantic partial credit, interpretation of free-text skills and GLM-5.3 CV extraction, which falls back to manual entry. The embedding model could also run locally; GLM-5.3 would need another host or model. Replit is essential only as the host of the live URL; the app could run on any Python host.

6. Security, Privacy & Responsible Use

6.1 Data Handling & Privacy

What user or personal data does your application collect, store, or send to third-party services (including AI APIs)? How is it protected, and how long is it kept? State "No user data is collected" if applicable.

[Maximum 150 words]

Demo and test data are synthetic. The app has no accounts. A student profile (skills, degree, semester, interests, projects, experience summary) is stored in SQLite with no name or contact field, and no expiry is set in code; whether the file survives Replit restarts is unverified. An uploaded CV PDF is read in memory only, never written to disk or logged; its text, cut to 12,000 characters, goes to GLM-5.3 on a RunPod Public Endpoint (retention under RunPod and Z.ai terms not verified), and the student must review the draft before matching. The embedding endpoint receives skill names and typed skill phrases, never whole profiles. GitHub queries return counts only. API keys are environment variables, masked in logs and not committed.

6.2 Known Limitations & Risks

Identify the main technical limitations of your AI components (e.g., hallucination, bias, rate limits, dependency on a single provider) and any mitigation you have implemented.

[Maximum 150 words]

CV extraction (GLM-5.3) can invent or misread skills. Mitigation: a skill is added only if its quote is found in the CV text, the student must review and confirm the draft, and a timeout falls back to manual entry. Semantic matching (0.60 threshold, calibrated on 52 fictional pairs) can map wrongly; mapped skills get half credit and a "please confirm" note, and known confusable pairs are blocked. The 20-skill catalogue is small for 30 roles, so rankings can be implausible, and four roles are flagged limited coverage. Trend signals are GitHub activity for 12 skills, not job demand, and move scores by at most 3 points. Both AI services depend on one provider, RunPod; if the embedding endpoint fails, results fall back to exact matching, labelled degraded. Cold starts take 20 to 25 seconds. Scores are not hiring predictions.

7. Compliance & Readiness Checklist

Tick all boxes that apply before submitting.

- ☑️ All APIs, cloud services, AI tools, models, and datasets used in the project are declared in this document.
- ☑️ All third-party code, datasets, and models are used in accordance with their licenses, and attribution is provided where required.
- ☑️ No API keys, passwords, or other credentials are exposed in the code repository, prototype, or this document.
- ☑️ Any use of generative-AI tools for development has been disclosed and reviewed by the team.
- ☑️ Any use of official SDC sponsor resources has been declared in Section 5.
- ☑️ The prototype described here is accessible to judges through the links provided in Section 1.
- ☑️ All fields in this template have been completed within the word limits stated.

8. Declaration

By submitting this document, the team leader confirms that the information provided is complete and accurate to the best of the team’s knowledge, and that every API, cloud service, AI tool, dataset, model, and sponsor resource used in the project has been declared.

Rehan Anjum (Team Leader)

25 October 2026